



Catchment Overview: River Llynfi long-term trends

A report prepared on behalf of Wye Salmon Association (WSA)

<i>Version</i>	<i>1.0</i>
<i>Date</i>	<i>18-April-2023</i>
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Contents

.....	1
Executive Summary	3
Introduction	4
Land use	5
Livestock.....	7
Phosphorus load modelling	7
Water quality trends	8
Llangorse Lake.....	8
Lower Llynfi.....	9
Dulas & Tryffrwd Brooks	10
River Ennig	12
Conclusion.....	13
References	14
List of abbreviations.....	14
Acknowledgements.....	14

Executive Summary

This work examines some long-term trends occurring within the Llynfi sub catchment in order to help understand the current position with respect to nutrient concentrations in surface water.

We start by examining land use data available from the UK Centre for Ecology and Hydrology. This shows that the sub catchment is dominated by grassland, with arable farming making up only 9% of the catchment land area. Moreover, in the period between 2000 and 2020, the fraction of improved grassland increased from 52% to 63%. This was achieved by the removal of a substantial area of broad-leaved woodland (4%) and the upgrading of some remaining rough pasture (5%). Over the same period, the area of arable land remained approximately constant, but appears to have consolidated in the lower catchment below Talgarth and in a few areas of the Dulas catchment. It would be expected that such an increase in improved grassland cover would increase phosphorus export to surface water.

We then examine livestock statistics for Wales and Powys in particular. These show a decline in populations of cattle (-11%) and sheep (-12%) over the same period. Such a fall in livestock numbers would serve to reduce phosphorus export. By applying the source apportionment model developed in our earlier report on the Llynfi (Green, 2022b), we show that this fall in export approximately balances the expected increase due to increased area of improved grassland. Thus, we would predict that surface water phosphorus concentrations should have remained fairly constant between 2000 and today.

Examining EA/NRW water quality data for the period 2000 to 2022 does indeed show constant levels of phosphorus and nitrate. We present data for Llangorse Lake, the Llynfi below the lake outlet and for the Llynfi above the Wye confluence, finding that all these locations show relatively constant nutrient concentrations over this period. The Dulas and Tryffrwd Brooks also show a similar pattern. The Ennig, over the same period shows a 32% increase in reactive phosphorus (RP) and a 17% fall in nitrate. The fall in nitrate is potentially explained by land use changes; however, the reason for the notable rise in RP is less obvious.

Overall, the surface water nutrient concentrations measured in the Llynfi sub catchment support the expectation of approximate stasis. The three Llynfi water bodies falling within the Wye SAC (Dulas, Tryffrwd & lower Llynfi) are all failing SAC targets for RP. The remaining water bodies (Ennig & middle Llynfi) are failing WFD targets for RP. The source apportionment model presented in our earlier work (Green, 2022b), estimated that over 80% of the phosphorus load comes from diffuse agricultural sources. Yet the only known remediation measure in the sub catchment during this period was the introduction of phosphate stripping at Talgarth WWTW by Dwr Cymru in 2015. If the failing SAC is to be remedied, it is clearly imperative that appropriate systemic changes are made within the livestock sector in order to drive down soil phosphorus levels and thereby reduce export to surface water.

Introduction

Wye Salmon Association (WSA) have published Catchment Overview reports on two sub catchments within the River Wye catchment, specifically the Llynfi and Garren sub catchments (Green, 2022b, 2022a). Whilst researching the Garren sub catchment, we discovered the benefit of an examination of long-term (~20-years) historical trends on water quality data recorded by the Environment Agency (EA). Trends on nitrate and orthophosphate in particular ran counter to the background of falling ruminant numbers over the period and instead correlated strongly with the rapid increase in poultry units within the sub catchment. The earlier Llynfi work did not include such an analysis. Whilst the Llynfi sub catchment does not contain poultry units, it may still be instructive to examine the equivalent data for this sub catchment. This, then, is the objective of this work.

Such a study is particularly germane in that the Llynfi sub catchment contains three water bodies (Dulas Brook, Tryffrwd Brook & lower Llynfi) falling within the River Wye SAC. Each of these water bodies fails its SAC target for reactive phosphorus. Upstream, Llangorse Lake, an SAC in its own right, is also failing reactive phosphorus targets. The remaining water bodies (Ennig and Llynfi between Llangorse and Dulas confluence) are failing to achieve WFD good status for reactive phosphorus.

The first stage of this study examines changes in land use over the period 2000 to 2020, based on digital mapping data downloaded from the UK Centre for Ecology & Hydrology (CEH) and analysed using QGIS software. We then add to this what can be gleaned about livestock populations over the period. The Garren work benefitted from detailed annual statistics for English counties (in this case Herefordshire) taken from the DEFRA website. Unfortunately, similar data for Powys does not appear to be readily available, and such data as is available does not go back before 2007. Nevertheless, we can get a general picture by combining statistics for the whole of Wales with what is known about Powys.

Using the source apportionment model developed in our earlier work, we then estimate the likely change in phosphorus export to surface water resulting from changes in land use and livestock numbers.

The second stage examines long-term water quality data for the Llynfi water bodies, which has been obtained from NRW via a freedom of information (FoI) request. This covers the period 2000 to 2023. In particular, we look at the trends on orthophosphate (and total phosphorus in the case of Llangorse Lake), as well as nitrate – both of which were informative in the Garren work.

Land use

Agricultural land use has been tracked over many years by CEH and is made available as GIS mapping data. The figures overleaf show the land use within the Llynfi sub catchment at two points in time. Figure 2 shows the situation in 2000; Figure 3 in 2020¹. It is clear that the Llynfi sub catchment is dominated by pasture. The relatively small fraction of arable farming occurs predominantly in the lower catchment. Moreover, it appears to have consolidated there in recent years. This perhaps reflects the general UK trend away from mixed farming to specialist livestock and arable operations.

The most striking difference between these maps is perhaps the marked reduction in broad-leaved woodland, a pattern which mirrors our findings on the Gamber. A close examination also shows that remaining areas of unimproved land are, by 2020, restricted to the upland areas. All lowland pasture has by then been improved.

We can readily summarise this data numerically and graphically, as follows.

Land use	2000	2020
Broad-leaved woodland	11%	7%
Coniferous woodland	1%	1%
Arable & horticulture	9%	9%
Improved grassland	53%	62%
Acid grass & other unimproved land	23%	19%
Freshwater	1%	1%
Urban & suburban	2%	2%

Table 1: land use changes between 2000 and 2020 within the Llynfi sub catchment (catchment above NRW Three Cocks gauging station)

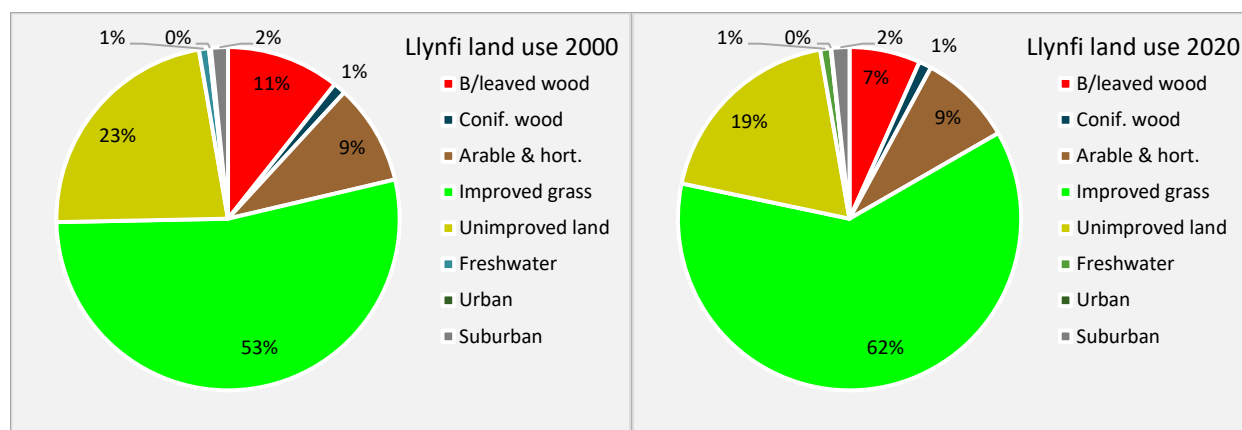


Figure 1: pie charts of land use in 2000 and 2020 within the Llynfi sub catchment

Thus, arable cropping has remained fairly constant over the last two decades, covering around 9% of the sub catchment. Over the same period, improved pasture has increased substantially, both by upgrading readily accessible rough land and by grubbing out almost half of the broad-leaved woodland. This is modern, intensive agriculture squeezing out every last drop.

¹ Note that more granular classifications were used in 2000, principally within the unimproved land types. However, urban, woodland, arable and improved grassland correspond directly.

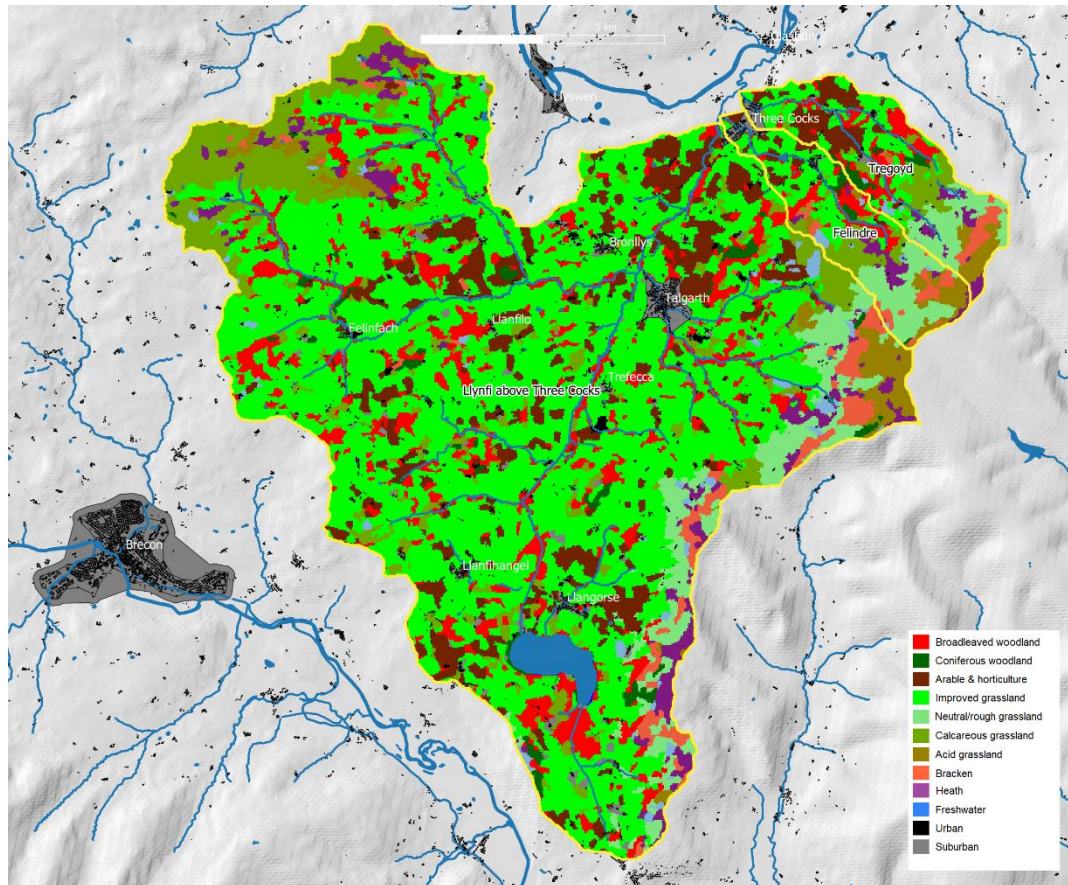


Figure 2: land use in 2000 within the Llynfi sub catchment

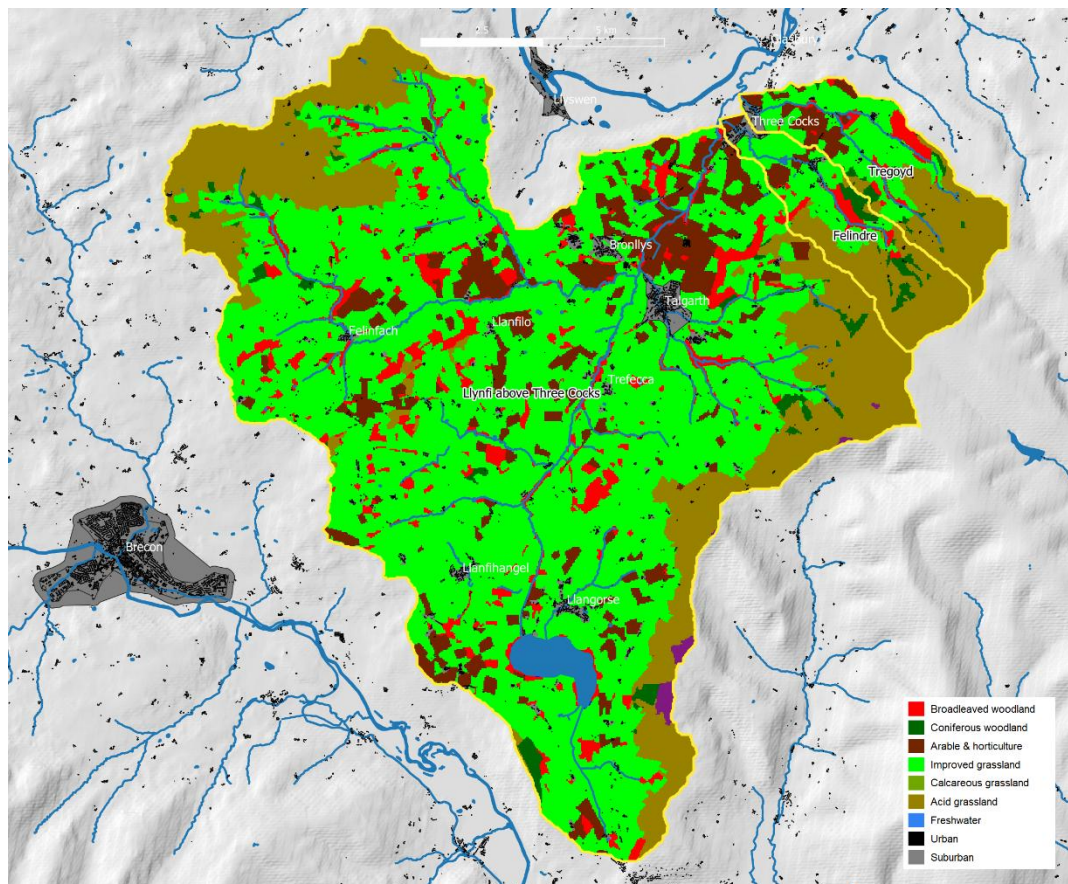


Figure 3: land use in 2020 within the Llynfi sub catchment

Livestock

Annual livestock statistics for the whole of Wales (<https://gov.wales/survey-agriculture-and-horticulture-june-2020>) show a long-term downward trend on both sheep and cattle numbers, as shown in the following charts.

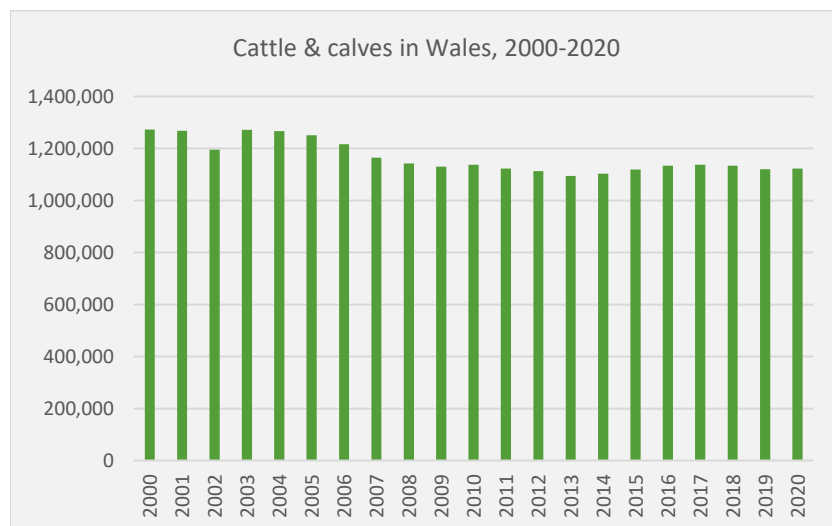


Figure 4: total cattle and calves in Wales over the period 2000 to 2020

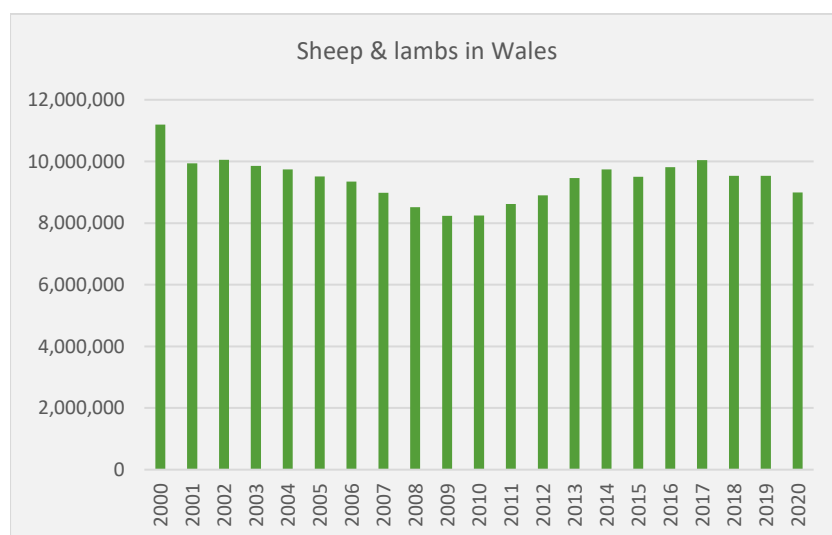


Figure 5: total sheep & lambs in Wales over the period 2000 to 2020

Sheep numbers peaked in Wales around 2000 and since then have declined by 12%. Cattle numbers peaked in the mid-1970s and have been in decline since then. Over the period 2000 – 2020, the cattle and calf population has fallen by 11%.

Numbers for Powys specifically are less readily available, but the [statswales website](#) has data for the period 2007 – 2017, which shows that Powys contained a consistent 18% of Welsh cattle and 39% of Welsh sheep throughout. Thus, it is reasonable to infer that Powys probably followed the national trend over the longer period of 2000 – 2020, with an approximate reduction of cattle and sheep populations of 11% & 12% respectively.

Phosphorus load modelling

In the preceding sections we have found (a) a 9% increase in improved grassland area which will drive an increase in phosphorus export to the river, and (b) an 11%-12% reduction in ruminant populations

which will tend to reduce the phosphorus export. We can (very roughly) model these effects using the source apportionment model which we developed in our earlier work on the Llynfi (Green, 2022b). In this, we estimated that 81% of the sub catchment phosphorus load was attributable to diffuse agricultural pollution, which in turn comprised 33% from improved grassland, 11% from cattle and 34% from sheep. Multiplying these numbers up suggests that the increase in improved grassland would be expected to increase river phosphorus loads by 2.4%. Conversely, the reduced ruminant numbers would be expected to reduce the phosphorus load by 4.1%. The nett effect is thus a 1.7% reduction over 20 years – not very much in the grand scheme of things and unlikely to be detectable.

Note however that this analysis takes no account of other phosphorus inputs. It ignores mineral fertiliser inputs to arable land (which may or may not have been appropriate to crop needs), or indeed alternative sources of organic fertiliser. Whilst poultry manure spreading is unlikely to be a significant factor in this sub catchment, there has been an increasing quantity of liquid digestate being spread, beginning around 2015. Much of this is going onto pasture and is highly unlikely to be in line with crop phosphorus needs, even if applicable nitrogen limits are being adhered to.

Water quality trends

In the following sections we present trend plots for two key nutrients, nitrate and phosphorus, both of which are driven by agricultural activities and, to a lesser extent, by sewage. We present phosphorus as *reactive phosphorus* (RP) which is reported by NRW as *orthophosphate, reactive*. The exception to this is Llangorse, where we present *total phosphorus* (TP), this being the compliance determinand for still waters.

Llangorse Lake

The long-term trend at Llangorse, as indicated by total phosphorus concentration, is one of constancy. Figure 6 below shows annual mean TP concentration from 2003 to 2022.

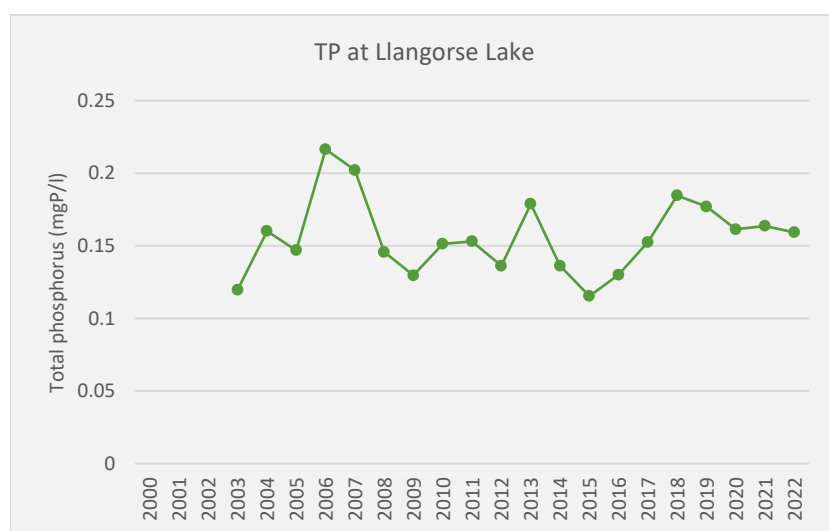


Figure 6: annual mean total phosphorus concentration measured in Llangorse Lake by EA/NRW over the period 2003 to 2022

It is clear that the values obtained over recent years, say 2018 to 2022, lie well within the spread of values found in 2002-2010. This pattern is replicated in the River Llynfi immediately below the lake outlet as shown in Figures 7 & 8. below. Both RP and nitrate show no obvious change over the period. Note that the significantly increased variance on the RP signal in recent years is the result of reduced sampling frequency due to budget cuts at NRW. RP at this point in the Llynfi has more variance than elsewhere in the Llynfi system, presumably the consequence of algal activity in the lake, and is thus difficult to interpret from 3 or 4 samples per year.

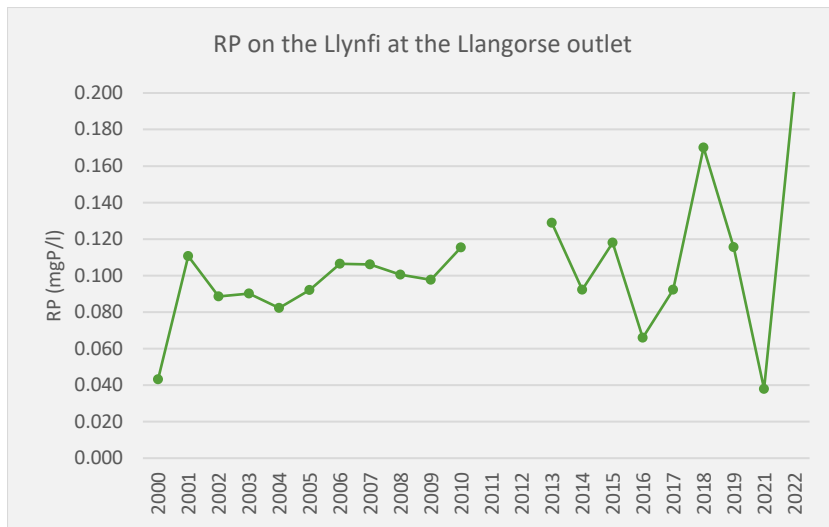


Figure 7: reactive phosphorus concentration measured in the Llynfi at the Llangorse outlet by EA/NRW over the period 2000 to 2022

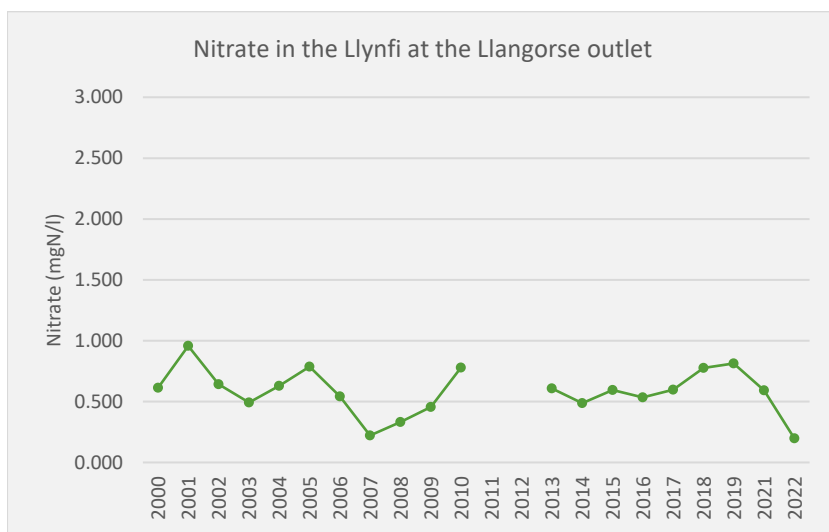


Figure 8: nitrate concentration measured in the Llynfi at the Llangorse outlet by EA/NRW over the period 2000 to 2022

Lower Llynfi

At the bottom of the Llynfi, annual mean RP concentration is significantly lower than up at Llangorse but otherwise shows similar long-term stability, as can be seen in Figure 9 below.

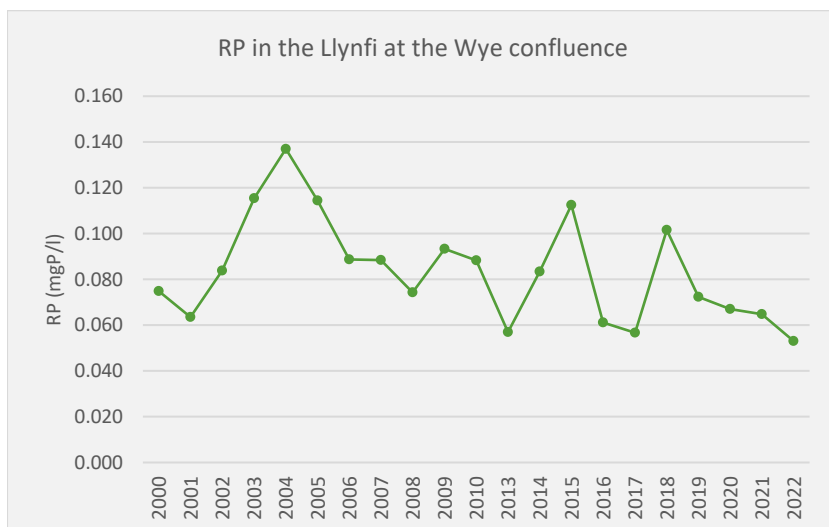


Figure 9: reactive phosphorus concentration measured in the Llynfi above the Wye confluence by EA/NRW over the period 2000 to 2022

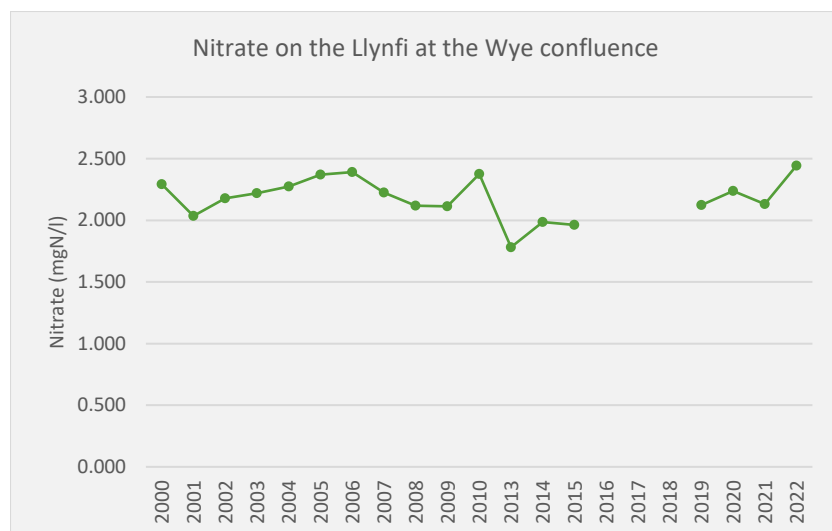


Figure 10: nitrate concentration measured in the Llynfi above the Wye confluence by EA/NRW over the period 2000 to 2022

Nitrate, likewise, has remained stable over the period, as is shown by Figure 10. As noted in our earlier work, mean nitrate values here are substantially higher than at Llangorse, possibly reflecting the concentration of arable farming in the lower catchment.

Dulas & Tryffrwd Brooks

Looking next at the major tributaries, we find that the Dulas also shows a stable long-term pattern for both RP and nitrate, shown in Figures 11 & 12. Once again, outliers for annual mean RP (2001 & 2019) are readily attributable to low sample counts that year.

Because it is an SAC water body, we also present the results for the Tryffrwd Brook in Figures 13 & 14. This also shows no long-term trend on either RP or nitrate signals.

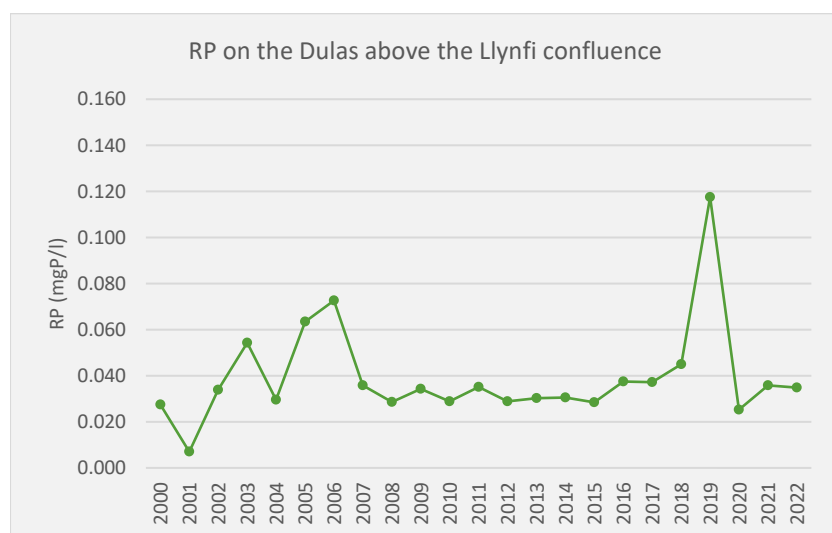


Figure 11: reactive phosphorus concentration measured in the Dulas above the Llynfi confluence by EA/NRW over the period 2000 to 2022

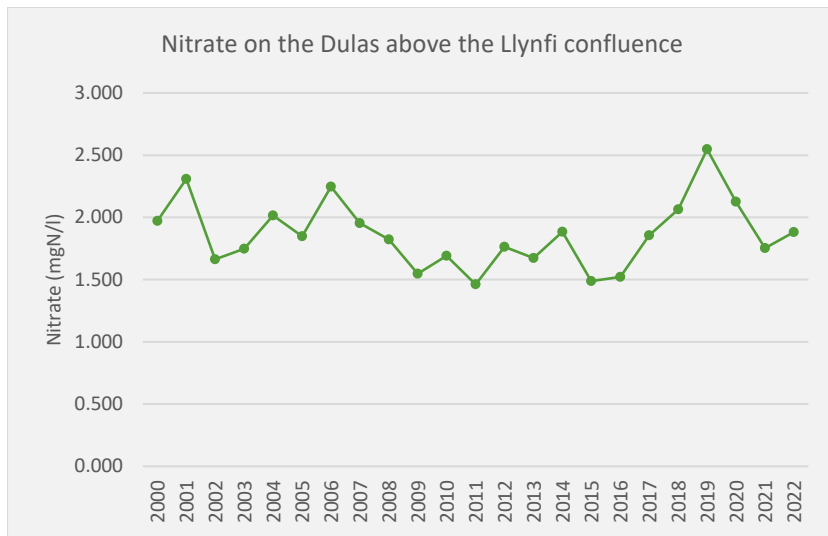


Figure 12: nitrate concentration measured in the Dulas above the Llynfi confluence by EA/NRW over the period 2000 to 2022

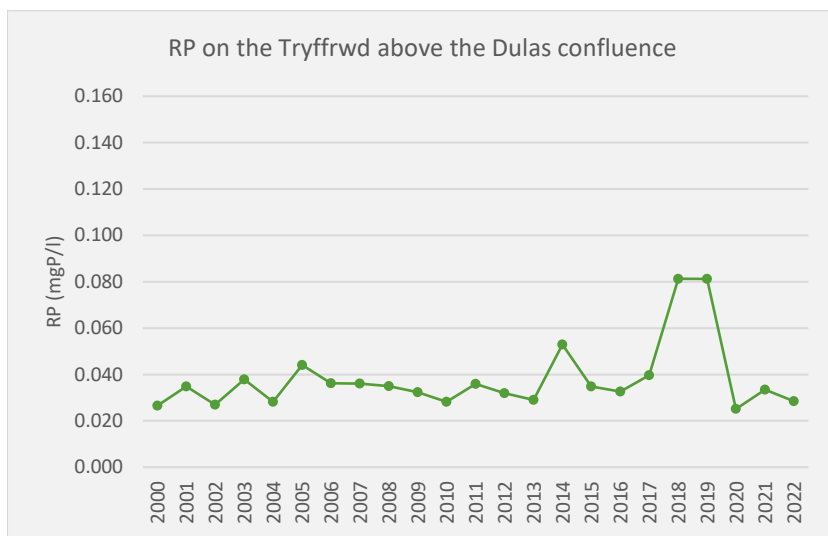


Figure 13: reactive phosphorus concentration measured in the Tryffrwd above the Dulas confluence by EA/NRW over the period 2000 to 2022

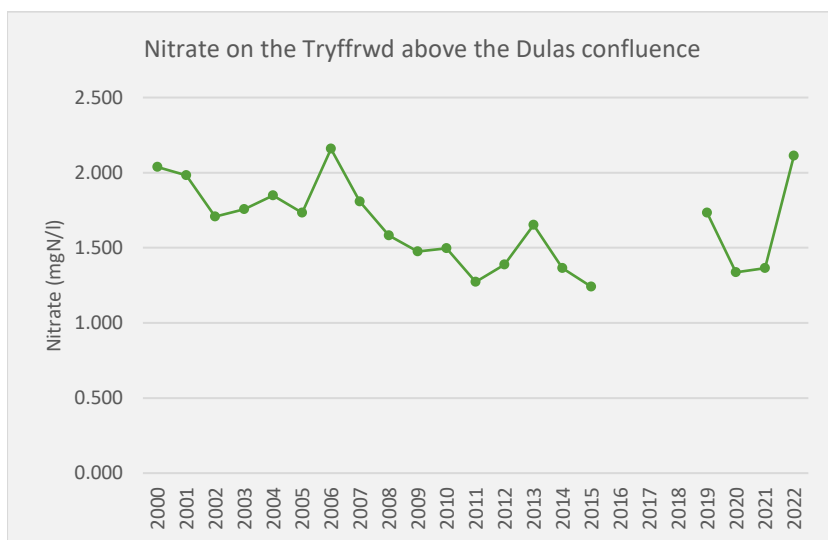


Figure 14: nitrate concentration measured in the Tryffrwd above the Dulas confluence by EA/NRW over the period 2000 to 2022

River Ennig

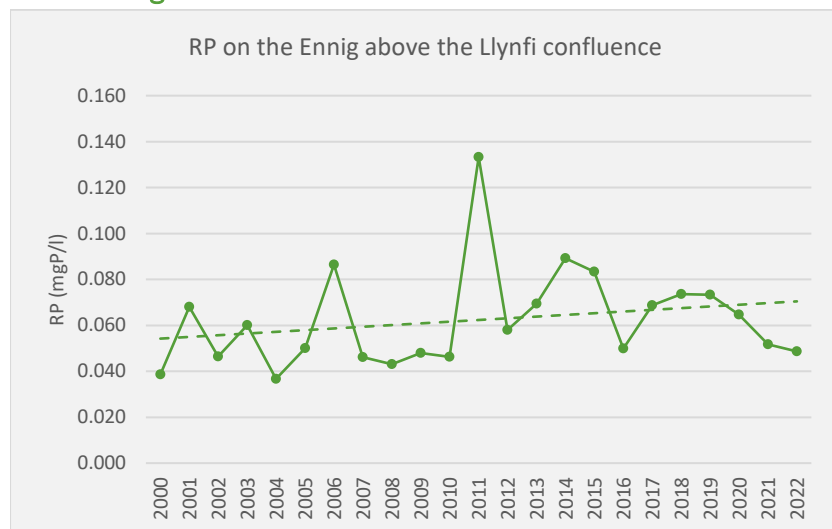


Figure 15: reactive phosphorus concentration measured in the Ennig above the Llynfi confluence by EA/NRW over the period 2000 to 2022

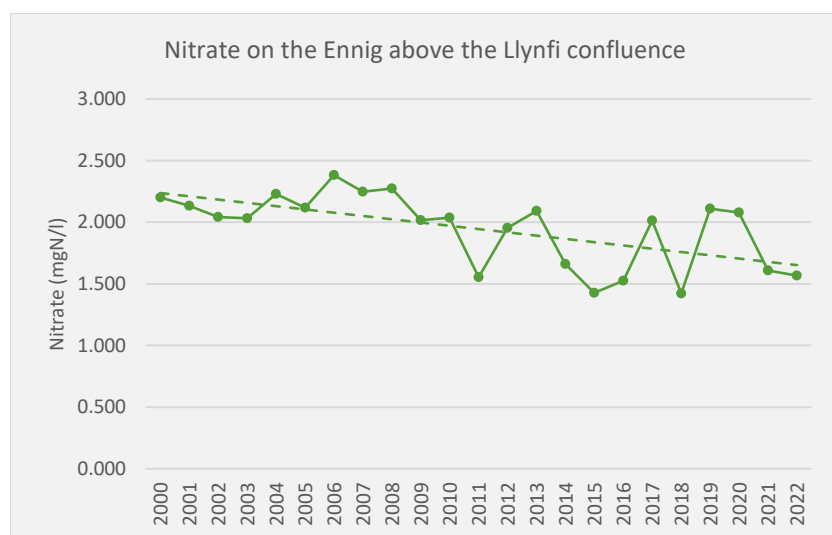


Figure 16: nitrate concentration measured in the Ennig above the Llynfi confluence by EA/NRW over the period 2000 to 2022

Only on the Ennig can we see the hint of some long-term change, on both RP and nitrate signals. The dashed lines in Figures 15 and 16 represent a regression line to the annual means. Since the annual means are based on variable sample sizes, we cannot infer anything quantitative from these regression lines, nevertheless they indicate that something is going on. Going back to the individual samples, we can take the mean values of the early years (2000-2005) and compare to the means values of the later years (2015-22). The results are shown in Table 2 below.

	RP (mgP/l)		Nitrate (mgN/l)	
	2000-05	2015-22	2000-05	2015-22
Samples	69	69	69	47
Mean	0.048	0.064	2.133	1.762
Change		+32%		-17%
t-test confidence		>99%		>99%

Table 2: Long-term changes in nutrient concentrations on the Ennig above the Llynfi confluence; t-test is double-sided.

Both RP and nitrate changes are highly statistically significant. Whilst reasons for the changes cannot be specifically attributed, the following circumstantial observations can be made. Arable land use in the Ennig catchment reduced by one fifth over the period (5% to 4% of catchment), which might account for the falling nitrate levels. At the same time, improved grassland increased - but no more so than on the Dulas and Tryffrwd. So, the cause of the increase in RP concentration on the Ennig is unclear.

Conclusion

In our study of the Garren (Green, 2022a), we found substantial rises in RP since 2000 which were attributable to manure spreading by the substantial concentration of intensive poultry units within that sub catchment. However, there are no known IPU's within the Llynfi sub catchment and nor are there any in the immediate vicinity. Whilst we cannot say that there is absolutely no poultry manure being spread within the Llynfi sub catchment, it is reasonable to expect, however, that this would not represent a major phosphorus input. Furthermore, the spreading of digestate is a relatively new phenomenon which, by analogy to Garren poultry, would not be expected to be visible in the rivers for several more years.

The Llynfi sub catchment is dominated by grassland. Between 2000 and 2020, improved grassland increased from 53% to 62% of sub catchment area. The increase was attributable to significant woodland clearance as well as conversion of some remaining unimproved land. The area of arable land (9%) remained constant, but activities became more concentrated over this period, possibly reflecting the general trend towards agricultural specialisation. Most arable farming now takes place in the lower catchment below Talgarth, but with a few significant pockets in the Dulas valley. Overall, therefore, a reasonable expectation would be to find RP concentrations increasing in response to the increase in improved grassland, but nitrate not changing a great deal.

However, over the same period, ruminant populations have fallen across Powys and, indeed, across Wales. Assuming these wider statistics to be representative of the Llynfi sub catchment, we expect a fall of 12% in sheep and 11% in cattle numbers over the period of interest. Such a stock reduction would tend to reduce the phosphorus export to the river.

Applying the source apportionment model developed in our first Llynfi report (Green, 2022b), suggests that the increase in phosphorus export due to increased area of improved pasture will approximately cancel the reduction expected due to falling livestock numbers. Thus, overall, we would predict no change in river RP and nitrate concentrations between 2000 and 2022.

And this is, indeed, what we find over most of the Llynfi catchment. Total phosphorus levels in Llangorse Lake have remained unchanged since 2000, as has both RP and nitrate concentrations immediately below the lake outlet. The Dulas and Tryffrwd have remained similarly constant, albeit with lower mean values, and the pattern repeats down at the Wye confluence. Only the Ennig shows any variation, having a statistically significant increase in RP concentration (+32%) and statistically significant fall in nitrate (-17%). Whilst the reduced nitrate could be reasonably explained by changes in land use, the increase in RP cannot. We might therefore suspect some specific local emissions are occurring.

It is clear that any mitigation measures implemented upstream of Llangorse Lake following the comprehensive 2008 study (May et al., 2008) have been largely ineffective. Downstream of the lake things are not much better. It would appear that, in a sub catchment containing three SAC waterbodies (since 2005), nothing of any significance has been done to address diffuse agricultural pollution. Our first Llynfi report suggested that this source accounts for over 80% of the phosphorus export to the rivers, yet the only identifiable improvement measure has been to introduce phosphate stripping at Talgarth WWTW in 2015. Further sewage treatment enhancements are planned in the AMP-8 investment cycle, but whilst these will help reduce RP concentrations during low flow conditions and are thus to be welcomed, they will have no impact on annual average statistics. In some senses, they act as a fig leaf to conceal inaction on the elephant in the room – agricultural nutrient leaching.

References

Green, G. (2022a). *Catchment Overview - Garren Brook*.

Green, G. (2022b). *Catchment Overview: River Llynfi*.

May, L., Dudley, B., Spears, B. M., & Hatton-Ellis, T. W. (2008). *Nutrient modelling and a nutrient budget for Llangorse Lake*.

List of abbreviations

CEH	UK Centre for Ecology & Hydrology
DC-WW	Dwr Cymru – Welsh Water
EA	Environment Agency
FoI	Freedom of Information
GIS	Geographical Information System (digital mapping)
LAM	Load Apportionment Modelling
NRW	Natural Resources Wales
RP	Reactive Phosphorus
SAC	Special Area of Conservation
TP	Total Phosphorus
WFD	Water Framework Directive

Acknowledgements

Catchment maps make use of OS data © crown copyright 2022, [OS Open Zoomstack | OS Products \(ordnancesurvey.co.uk\)](#) and are prepared and analysed using [QGIS](#) open-source software.

Land use data is taken from CEH digital mapping.

Water quality data has been provided by NRW via an FoI request.